

Sharing Into Equal Groups — How Many Groups? How Many in Each?

Answer Key

Grades 2–3

Quick Answers

1. 6	3. 6	5. 6	7. 9
2. 4	4. 7	6. 5	8. 5

Curriculum

Level: Grades 2–3

3.OA.A.3 / 4.OA.A.2 – problems 1, 2, 3, 4, 6, 7, 8

3.OA.C.7 – problem 5

Difficulty 1–5 of 5 across 9 tagged checks.

Common wrong answers

1. *If they got 3: answered with the number of bags, which is the group count, not how many stickers are in each bag*
 2. *If they got 5: answered with the number in each bag, which is the group size, not the number of bags*
 3. *If they got 4: repeated the crackers-on-each-plate number as the number of plates*
 4. *If they got 5: repeated the number of pots as the number of seeds in each pot*
 6. *If they got 8: repeated the markers-per-group number as the number of groups*
 7. *If they got 4: gave the number of rows instead of the shells in one row, which is the other child's story*
 8. *If they got 6: repeated the pencils-in-each-cup number as the number of cups*
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1. 18 stickers into 3 bags. How many in each bag?

Step 1. Sort the numbers by their job. The total is 18. The story gives the number of *groups*: 3 bags. The question asks for the size of each group.

Step 2. Share the total into the groups: $18 \div 3$. Skip-count by 3 to check: 3, 6, 9, 12, 15, 18 — six threes.

Step 3. Each bag gets 6 stickers.

Step 4. Multiply back: 3 bags $\times$ 6 in each bag = 18 ✓.

Watch for: an answer of 3. That is the number of bags, which the story already told us — it cannot also be the answer.

2. 20 stickers, 5 in each bag. How many bags?

Step 1. Sort the numbers. Total = 20. This time the story gives the size of each group (5 in a bag), and the question asks how many groups.

Step 2. Keep taking 5 away, or skip-count by 5: 5, 10, 15, 20. That is four fives.

Step 3. Mia fills $\boxed{4}$ bags.

Step 4. Multiply back: 4 bags $\times$ 5 in each bag = 20 $\checkmark$.

Watch for: an answer of 5. That is how many go in one bag, not how many bags there are.

3. Ravi: 24 crackers, 4 on each plate — he says 4 plates.

Step 1. Find the number Ravi wrote back. The story said “4 crackers on a plate,” so 4 is the *size of each group*. Ravi used it again as the number of plates.

Step 2. The question asks how many plates, so share the total into groups of four: $24 \div 4$. Skip-count by 4: 4, 8, 12, 16, 20, 24 — six fours.

Step 3. There are really $\boxed{6}$ plates.

Step 4. Multiply back: 6 plates $\times$ 4 crackers = 24 $\checkmark$. Ravi’s answer would need $4 \times 4 = 16$ crackers, which is not the total the story gave.

4. Leo: 35 seeds in 5 pots — he says 5 seeds in each pot.

Step 1. Find the repeat. The story said 5 *pots*, so 5 is the number of groups. Leo used that same number for the seeds in each pot.

Step 2. The question asks for the size of one group: $35 \div 5$. Skip-count by 5 to 35: 5, 10, 15, 20, 25, 30, 35 — seven fives.

Step 3. Each pot gets $\boxed{7}$ seeds.

Step 4. Multiply back: 5 pots $\times$ 7 seeds = 35 $\checkmark$. Leo’s answer would only use $5 \times 5 = 25$ seeds, leaving ten in his hand — a good way for a child to see the error without being told.

5. Ana: 48 apples, 8 in each basket. Fill in $\underline{\hspace{1cm}}$ $\times$ 8 = 48.

Step 1. Read the equation in story words: (number of baskets) $\times$ (8 apples in each basket) = 48 apples in all. The blank is the number of groups.

Step 2. Count up by eights until you reach the total: 8, 16, 24, 32, 40, 48. That is six eights.

Step 3. The missing number is $\boxed{6}$, so Ana fills six baskets.

Step 4. Check: $6 \times 8 = 48$ $\checkmark$. Writing the multiplication first is the surest cure for this mix-up — the blank sits in exactly one of the two jobs, so it cannot be confused with the other.

6. Rio: 40 markers, 8 to each group — he says 8 groups.

Step 1. Check Rio’s answer the way the problem asks. If there were 8 groups with 8 markers each, the teacher would need $8 \times 8 = 64$ markers. That is bigger than 40, so Rio’s answer cannot be right.

Step 2. Rio repeated the group size (8 markers to each group) as the number of groups.

Step 3. Share the total into groups of eight: $40 \div 8$. Skip-count: 8, 16, 24, 32, 40 — five eights.

Step 4. Five groups get markers: $\boxed{5}$, and the check is $5 \times 8 = 40$ $\checkmark$.

7. Jen has 36 shells with 4 in each row; Ben has 36 shells in 4 rows. How many shells in one of Ben’s rows?

Step 1. The two stories use the same two numbers, but 4 has a different job in each. For Jen, 4 is the size of each row. For Ben, 4 is the number of rows.

Step 2. The question is about Ben, so 4 is a group count and the missing number is the group size: $36 \div 4$.

Step 3. Skip-count by 4 to 36: 4, 8, 12, 16, 20, 24, 28, 32, 36 — nine fours. Ben has 9 shells in each row.

Step 4. Multiply back: 4 rows $\times$ 9 shells = 36 $\checkmark$. Jen’s rows hold only 4 shells each, so answering 4 here means answering Jen’s story instead of Ben’s.

8. (a) How do you tell the two jobs apart? (b) 30 pencils, 6 in each cup — how many cups?

(a) *Open response — flagged for manual review. Model answer below; accept any equivalent wording.*

Look at the words right next to the number. “6 cups” counts the groups. “6 in each cup” tells the size of one group — the words “in each” are the giveaway. The biggest number is usually the total, the amount being shared out. Then ask what the question wants, and remember that the answer is never a number the story already gave you.

What a full-credit answer must contain: the “in each” test (or an equivalent description of it), and the idea that the number the story gives cannot also be the answer.

(b) Here 6 comes with the words “in each cup,” so it is the group size, and the question asks for the number of groups.

Step 1. Share the total into sixes: $30 \div 6$.

Step 2. Skip-count by 6: 6, 12, 18, 24, 30 — five sixes.

Step 3. Five cups are filled: 5, checked by $5 \times 6 = 30 \checkmark$.